

5.1 -Task3: Deploying a Cloud Instance

Deploying a cloud environment to launch your cyber security app training!

In this exercise, you'll use a cloud interface similar to the AWS platform. The goal is not to memorize buttons, but to understand how cloud resources are easily created and managed in a real-world scenario.

Basic Cloud Terminology

To complete this exercise, you only need to understand a few basic concepts from AWS:

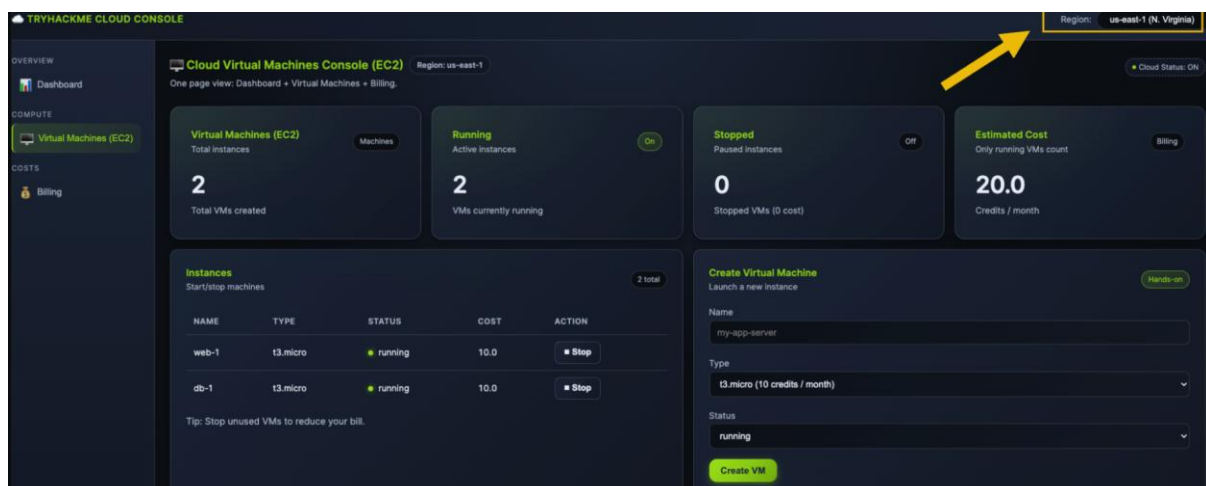
- **EC2 (Virtual Computer / Server):** EC2 represents a virtual computer in the cloud. Just like a real computer, it has a CPU and memory (RAM) and can run applications. Whenever you add an EC2 instance, you are adding a computer to your environment.
- **Instance Type (for example: t2, t3, m5):** Instance types describe how powerful the virtual computer is. Some have more CPU and RAM and are therefore more expensive. You choose the Instance Type based on your needs, knowing that:
 - Bigger instances = more power + higher cost
 - Minor instances = less power + lower cost

Deploying Your Environment

You will create three virtual computers (EC2 instances) to host your cyber security training application. This aligns with the Infrastructure as a Service (IaaS) model you previously learned, as cyber security practices often require full access to the operating system. This allows you to install tools, configure the system, and safely simulate attacks and defenses, just like in real-world environments!

Picking a Region

First, you need to choose a region where your resources will live. You can do it in the top right of your screen:



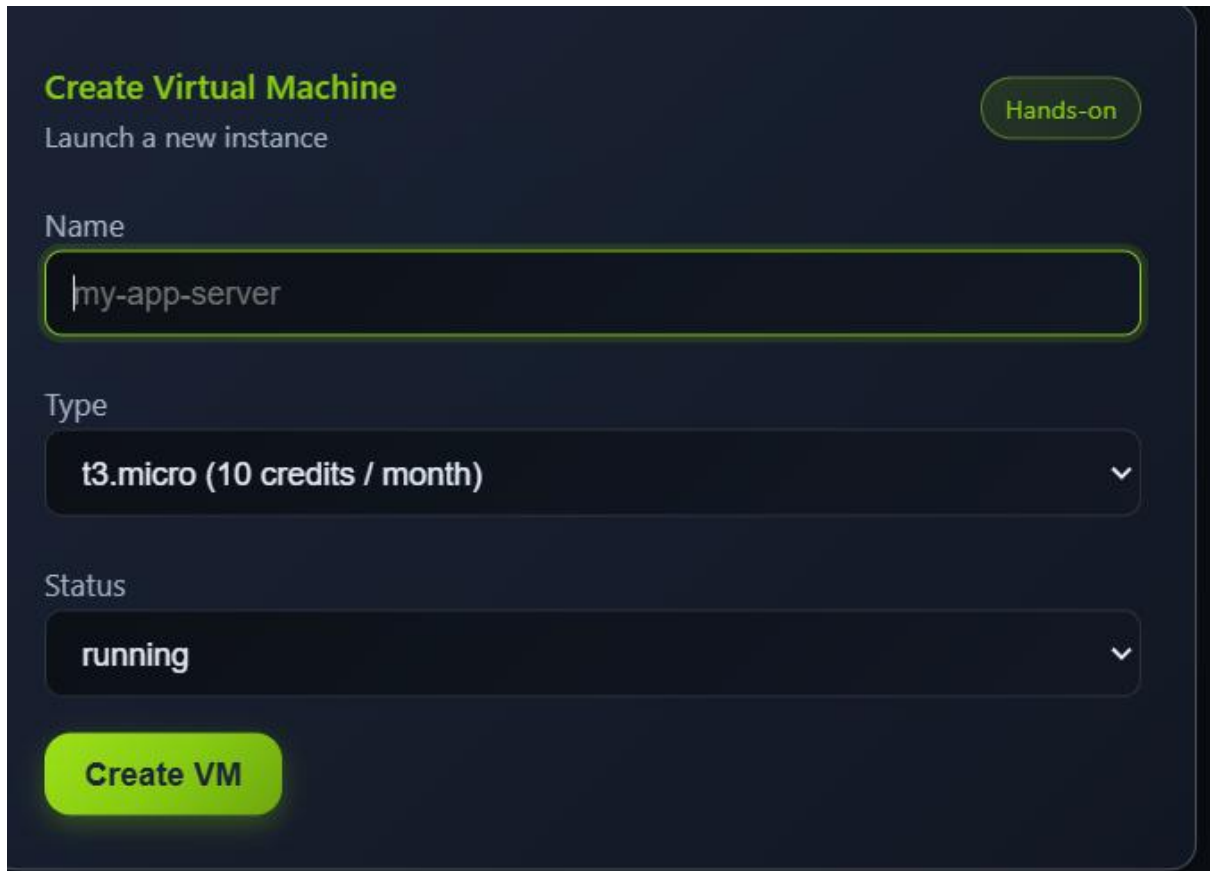
A region represents a geographical location, for example, Europe or North America.

Creating Lab Machines

Now, go to the Create Lab Machine block on the right side of the page to create the lab machines for your application.

First, let's create your application interface machine. Set the following configuration:

- Instance Name: application-interface
- Instance Type: t3.micro
- Status: running



The screenshot shows a 'Create Virtual Machine' form with a dark blue background. At the top left, the title 'Create Virtual Machine' is in green, with the subtitle 'Launch a new instance' below it. A green 'Hands-on' button is in the top right. The form has three input fields: 'Name' with the text 'my-app-server', 'Type' with a dropdown menu showing 't3.micro (10 credits / month)', and 'Status' with a dropdown menu showing 'running'. A green 'Create VM' button is at the bottom left.

Now, let's create two testing computers for users to practice their cyber security skills.

Since these are test machines, let's use a more powerful instance type: m5.large.

Machine 1:

- Instance Name: study-machine-1
- Instance Type: m5.large
- Status: running

Machine 2:

- Instance Name: study-machine-2

- Instance Type: m5.large
- Status: running

Create Virtual Machine

Launch a new instance

Hands-on

Name

study-machine-1

Type

m5.large (70 credits / month) ▼

Status

running ▼

Create VM

Create Virtual Machine

Launch a new instance

Hands-on

Name

study-machine-2

Type

m5.large (70 credits / month) ▼

Status

running ▼

Create VM

Instances

5 total

Start/stop machines

NAME	TYPE	STATUS	COST	ACTION
web-1	t3.micro	● running	10.0	■ Stop
db-1	t3.micro	● running	10.0	■ Stop
application-interface	t3.micro	● running	10.0	■ Stop
study-machine-1	m5.large	● running	70.0	■ Stop
study-machine-2	m5.large	● running	70.0	■ Stop

Tip: Stop unused VMs to reduce your bill.

Billing Analysis

Let's analyze how much credit is costing our environment by navigating to the Billing section at the bottom of the page.

There, you can check how much each type of instance is costing you, as well as the total.

Only running VMs add cost

NAME	TYPE	STATUS	MONTHLY COST
web-1	t3.micro	running	10.0
db-1	t3.micro	running	10.0
application-interface	t3.micro	running	10.0
study-machine-1	m5.large	running	70.0
study-machine-2	m5.large	running	70.0

Total estimated: **170.0** credits / month

Currently, you are still developing your application, so users have not yet accessed your platform. We can optimize costs by stopping the two study machines and reviewing the new cost of your environment.

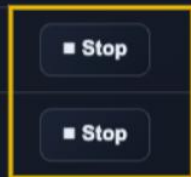
Go to the Instances block and click the Stop button for both study-machine-1 and study-machine-2.

Instances5 total

Start/stop machines

NAME	TYPE	STATUS	COST	ACTION
web-1	t3.micro	● running	10.0	<button>■ Stop</button>
db-1	t3.micro	● running	10.0	<button>■ Stop</button>
application-interface	t3.micro	● running	10.0	<button>■ Stop</button>
study-machine1	m5.large	● running	70.0	<button>■ Stop</button>
study-machine2	m5.large	● running	70.0	<button>■ Stop</button>

Tip: Stop unused VMs to reduce your bill.



Go back to the Billing section, and you should see that your cost was reduced considerably:

Billing

Summary

Only running VMs add cost

NAME	TYPE	STATUS	MONTHLY COST
web-1	t3.micro	running	10.0
db-1	t3.micro	running	10.0
application-interface	t3.micro	running	10.0
study-machine-1	m5.large	stopped	0.0
study-machine-2	m5.large	stopped	0.0

Total estimated: 30.0 credits / month

This exercise is a clear example of how easily you can change and manage your costs in a cloud environment.

Answer the following questions to conclude your learning about cloud computing!

Answer the questions below

What is the total cost of credits of the entire environment if study-machine-1 and study-machine-2 are stopped?

30

How many credits does an m5.large EC2 instance cost per month?

70

What is the total cost of credits if only the new instances we created are running?

Pricing reference

Credits / month (running only)

Reference

- **t3.micro** → 10 credits
- **t3.small** → 18 credits
- **m5.large** → 70 credits
- **Stopped VM** → 0 credits

150

What would be the total running cost of the entire environment you created if you add a third t3a.small study machine?

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In this room, you explored the core ideas behind cloud computing and how it enables flexible, scalable, and cost-effective access to computing resources.

Key Terminology

Let's quickly review some concepts we learned in this room:

- **Public Cloud**
Cloud services you access over the internet that many people and companies share.
- **Private Cloud**
A cloud built just for one company, so they have more control and security.
- **Hybrid Cloud**
A mix of public and private clouds that can work together and share data.
- **IaaS**
A service where you rent basic computer parts like servers and storage from the cloud.
- **PaaS**
A service that gives you a ready-to-use environment to build and run apps without managing servers.
- **SaaS**
Software you use online without installing anything, like Gmail or Zoom.
- **EC2**
Amazon's cloud computers that you can quickly create, use, and resize whenever you need them.

We also concluded that the key benefits of cloud computing are:

- **Scalability**
- **On-demand self-service**
- **Pay only for what you use**
- **Security**
- **High availability**
- **Global access**